Quantitative Genetic Models

Lectures 1 – 4 **Prof Georgios Banos** **SRUC** **Introduction to quantitative (linear) models**	**Lectures 5 – 10** **Prof John Woolliams (course leader)** **Roslin-UoE** **Quantitative models with genomic data**
Computer Practicals **Demonstrator(s)**	

Leading the way in Agriculture and Rural Research, Education and Consulting

By the end of Lecture 4

- Mixed model
  - Random vs. fixed effects
- Building a quantitative (genetic) model
- Effect estimation
  - Genetic effects (genetic evaluation/prediction)
- Variance de-composition
  - (Genetic) parameter estimation (e.g. trait heritability)
- Model utility

Applicability

SRUC

Teaching staff mainly from Livestock Genetics but ….

…. principles also apply to:

- **Human (Medical) Complex Trait Genetics**
- **Evolutionary Genetics (Biology)**
- **Ecological Genetics**
- **Plant Sciences**
- **Forestry**

Study of quantitative traits

- **Quantitative trait:**

- **Qualitative trait:**

Study of quantitative traits

- **Quantitative trait:**
  - Many genes (polygenic)
  - Environment
  - Gene by environment interaction

- **Qualitative trait:**
  - Single or few genes (monogenic – oligogenic)

Study of quantitative traits

- Quantitative trait (examples):
  - Obesity, schizophrenia, cancer
  - Livestock traits: milk yield, egg production, growth, fertility
  - Disease resistance
    - Most diseases of interest in humans, animals and plants
- Qualitative trait (examples):
  - Eye/hair colour
  - Some diseases (genetic defects)
    - Haemophilia
    - Transmissible Spongiform Encephalopathy (sheep/goats)

Modelling life

Live body weight in beef cattle

What influences it?

$$y = X\beta + Zu + \epsilon$$

$$\begin{pmatrix} X'R^{-1}X & X'R^{-1}Z \\ Z'R^{-1}X & Z'R^{-1}Z + G^{-1} \end{pmatrix} \begin{pmatrix} \tilde{\beta} \\ \tilde{u} \end{pmatrix} = \begin{pmatrix} X'R^{-1}y \\ Z'R^{-1}y \end{pmatrix}$$

Random vs. fixed effects

- Random
  - Sampled from a (normal) distribution
  - All levels are not represented in study
  - Individual (Genetic) effect
    - i.e. a sample of all individuals in the population are included
  - Sometimes Contemporary Group (e.g. cohort, herd) effect
- Fixed
  - Non-random
  - All (most) levels represented in study
  - Discrete categories (e.g. gender, calendar season, treatment)
    - Class variables, discreet variables, factor variables
  - Continuous scale (e.g. age)
    - Covariates

- A <u>mixed model</u> includes <u>both</u> fixed and random effects

Basic model

Phenotype = Genotype + “Environment”

$$P = G + E$$

Differences (variation) observed in P are due to differences (variation) in G and in E

$$Var(P) = Var(G) + Var(E)$$

Basic model

Phenotype = Genotype + "Environment"

$$P = G + E$$

$$P = A + D + I + E$$

$$Var(P) = Var(A) + Var(D) + Var(I) + Var(E)$$

Inherited

Coefficient of heritability (h^2) = Var(A) / Var(P)
Specific for each trait
Range: 0.0 – 1.0

Basic problem

- Decomposing the sources of variation that contribute to observed phenotypes
- P = Genetic (G) + Environment (E)
- Thus Var(P) = Var(G) + Var(E)
  - Sources of Var(E): environmental (e.g. season, climate), management factors (e.g. production stage, feeding ration, treatment), individual non-genetic factors (e.g. age), other non-genetic factors
  - Sources of Var(G): different forms of inheritance leading to different components of genetic variance (e.g. additive Var(G), dominance Var(G), epistatic Var(G))

Basic problem

Accurate assessment and estimation of

- Var(G) ➔ variance components ➔ genetic parameters
- G ➔ genotype ➔ genetic values ➔ genetic evaluation

involves identifying and accounting for non-genetic systematic sources of variation (Var(E))

Accuracy of estimation increases with amount of information (data) available

- Number of individual records
- Number of records on relatives

Linear mixed model

- Provides a framework to assess factors affecting phenotypic observations
- Consists of
  - Dependent/response variable: The phenotypic data vector (y) which is a set of observations on the study units (traits of humans, livestock, wild animals, trees, plants)
    - Often assuming that data follow a normal distribution ~N(μ, V)
  - Independent/explanatory variables: Factors which may have an effect on the observations
    - Random/fixed
    - Central to the analysis (e.g. a treatment effect)
    - Nuisance/noise factors (expected to have an effect on the trait of study and are included to improve accuracy and avoid bias)
    - Category effects (class/discreet variables ordered 1 to p)
    - Covariates (continuous variables, centred)

Linear fixed effects model

- Dependent variable: cow milk yield (amount of milk produced by a cow in a defined time interval)
- Independent variables: herd, season
- Simple linear model (not necessarily complete):

$$y_{ijk} = \mu + h_i + c_j + \varepsilon_{ijk}$$

where

y_{ijk} = milk yield of the kth cow in i^{th} herd and j^{th} season

μ = overall population mean

h_i = effect of the ith herd (i=1 …. p_h)

c_j = effect of the jth season (j=1 …. p_c)

ε_{ijk} = residual effect

Matrix form

- Suppose there are 4 cows, 2 herds and 2 seasons:

$$y_{111} = \mu + h_1 + c_1 + \varepsilon_{111}$$
$$y_{212} = \mu + h_1 + c_2 + \varepsilon_{212}$$
$$y_{321} = \mu + h_2 + c_1 + \varepsilon_{321}$$
$$y_{422} = \mu + h_2 + c_2 + \varepsilon_{422}$$

$$X = \begin{array}{c} \begin{array}{cccc} h_1 & h_2 & c_1 & c_2 \end{array} \\ \begin{pmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix} \end{array}$$

- In matrix form previous equation can be written as:

$$y = Xb + e$$

y = n x 1 vector of observations; n = number of records (4)

b = p x 1 vector of fixed effects; p = number of levels for fixed effects (4)

e = n x 1 vector of random residual effects

X = matrix of order n x p (4 x 4), that relates records to fixed effects

X = <u>incidence matrix</u> as it describes the 'incidence' of the different fixed effects in the observed data; also called a <u>design matrix</u> as it encapsulates the design of the study.

Linear mixed model

$$y = Xb + Zu + e$$

where

- y = *n* x 1 vector of observations; *n* = number of records
- b = *p* x 1 vector of fixed effects; *p* = number of levels for fixed effects
- X = design matrix of order *n* x *p*, that relates records to fixed effects
- u = *q* x 1 vector of random (individual, genetic) effects; *q* = number of levels for random effects
- e = *n* x 1 vector of random residual effects
- Z = design matrix of order *n* x *q*, that relates records to random individual effects

Linear mixed model - facts

- With random effects in the model, solutions will depend on the variance estimates of the random effects

- Random effects are usually assumed to be drawn from a distribution N(0, V)

- For genetic effects in a basic mixed linear model, this assumption is equivalent to assuming that traits are determined by additive alleles, each with a small effect, at infinitely many unlinked loci. This is called the infinitesimal model

Linear mixed model - assumptions

Expectations (E) of the variables are

E(y) = Xb

E(u) = E(e) = 0 ➔

$$E\begin{pmatrix} \mathbf{y} \\ \boldsymbol{u} \\ \mathbf{e} \end{pmatrix} = \begin{pmatrix} \mathbf{Xb} \\ \mathbf{0} \\ \mathbf{0} \end{pmatrix}$$

Var(e) = R = $I\sigma^2_e$

Var(u) = G = $I\sigma^2_u$ or $A\sigma^2_u$

Cov(u, e) = Cov(e, u) = 0

$$V\begin{pmatrix} \boldsymbol{u} \\ \mathbf{e} \end{pmatrix} = \begin{pmatrix} \mathbf{G} & \mathbf{0} \\ \mathbf{0} & \mathbf{R} \end{pmatrix}$$

Linear mixed model - assumptions

$$\mathrm{Var}(y) = V = \mathrm{Var}(Xb + Zu + e)$$
$$= Z\,\mathrm{Var}(u)\,Z' + \mathrm{Var}(e) + \mathrm{Cov}(Zu, e) + \mathrm{Cov}(e, Zu)$$
$$= ZGZ' + R + Z\,\mathrm{Cov}(u, e) + \mathrm{Cov}(e, u)Z'$$

Since $\mathrm{Cov}(u, e) = \mathrm{Cov}(e, u) = 0$, then

$$V = ZGZ' + R$$

$$\mathrm{Cov}(y, u) = \mathrm{Cov}(Xb + Zu + e, u)$$
$$= \mathrm{Cov}(Zu, u) + \mathrm{Cov}(e, u)$$
$$= Z\,\mathrm{Cov}(u, u)$$
$$= ZG$$

$$\mathrm{Cov}(y, e) = \mathrm{Cov}(Xb + Zu + e, e)$$
$$= \mathrm{Cov}(Zu, e) + \mathrm{Cov}(e, e)$$
$$= Z\,\mathrm{Cov}(u, e) + \mathrm{Cov}(e, e)$$
$$= R$$

Estimation and prediction

- From the linear mixed model we want to
  - Estimate *(a linear function of)* the fixed effects
  - Predict *(a linear function of)* the random effects

- Fixed effects (b)
  - Best Linear Unbiased Estimator (BLUE)
    - Best: maximises the correlation between true (unknown) and estimated values
    - Linear: estimates are linear functions of the data
    - Unbiased: estimates are such that $E(\mathbf{b} \mid \hat{\mathbf{b}}) = \mathbf{b}$
  - $b = (X'V^{-1}X)X'V^{-1}y$
  - generalised least square solution
  - k' b (contrasts)

Estimation and prediction

- Random effects (u):
  - Best Linear Unbiased Predictor (BLUP)
  - $\hat{u}$ = BLUP (u) = $GZ'V^{-1}(y-Xb)$
    - Adjusted for all fixed effects in the model
  - Requires estimates of G=Var(u)

  - Estimation of genetic effect (genetic evaluation)
    - Estimate the overall genetic merit of individuals with regards to a trait/characteristic
    - Estimated Breeding Value (EBV)
    - Predicted Transmitting Ability (PTA)
  - Estimation of gene effects

Mixed Model Equations

- The previous equation for the BLUP of a random effect require V^{-1}, which is not always computationally feasible with large data
- Mixed Model Equations (MME) estimate b and u <u>simultaneously</u> without the need for computing V^{-1}. The MME are:

$$\begin{bmatrix} X'R^{-1}X & X'R^{-1}Z \\ Z'R^{-1}X & Z'R^{-1}Z+G^{-1} \end{bmatrix} * \begin{bmatrix} \hat{b} \\ \hat{u} \end{bmatrix} = \begin{bmatrix} X'R^{-1}y \\ Z'R^{-1}y \end{bmatrix}$$

MME

$R = I\sigma^2_e$ $\qquad$ $G = A\sigma^2_u$

Then MME:

$$\begin{bmatrix} \mathbf{X'X} & \mathbf{X'Z} \\ \mathbf{Z'X} & \mathbf{Z'Z} + A^{-1}\lambda \end{bmatrix} * \begin{bmatrix} \hat{\mathbf{b}} \\ \hat{\mathbf{u}} \end{bmatrix} = \begin{bmatrix} \mathbf{X'y} \\ \mathbf{Z'y} \end{bmatrix}$$

Where:

$\lambda = \sigma^2_e/\sigma^2_u$ (function of trait heritability)

A = matrix of genetic relationships among individuals (next lecture, tutorials)

Example Data

Cow	Herd	Season	Milk Yield (kg/d)
1	1	2	36.2
2	1	1	25.8
3	1	2	31.5
4	1	1	42.0
5	1	2	12.3
6	2	2	28.5
7	2	1	10.6
8	2	2	23.4
9	2	2	22.4
10	2	1	14.8

y

Transpose of $X'y = \begin{pmatrix} 147.8 & 99.7 & 93.2 & 154.3 \end{pmatrix}$

$$X' = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 \\ \hline 0 & 1 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 & 1 & 0 & 1 & 1 & 0 \end{pmatrix}$$

$$X'X = \begin{pmatrix} \begin{matrix} 5 & 0 \\ 0 & 5 \end{matrix} & \begin{matrix} 2 & 3 \\ 2 & 3 \end{matrix} \\ \begin{matrix} 2 & 2 \\ 3 & 3 \end{matrix} & \begin{matrix} 4 & 0 \\ 0 & 6 \end{matrix} \end{pmatrix}$$

Example MME

$$\begin{bmatrix} \boxed{X'X} & X'Z \\ Z'X & Z'Z + A^{-1}\lambda \end{bmatrix} * \begin{bmatrix} \boxed{\hat{b}} \\ \hat{u} \end{bmatrix} = \begin{bmatrix} \boxed{X'y} \\ Z'y \end{bmatrix} \qquad \begin{bmatrix} X'X & \\ & \end{bmatrix} * \begin{bmatrix} \hat{b} \\ \end{bmatrix} = \begin{bmatrix} X'y \\ \end{bmatrix}$$

$$\begin{pmatrix} 5 & 0 & 2 & 3 \\ 0 & 5 & 2 & 3 \\ 2 & 2 & 4 & 0 \\ 3 & 3 & 0 & 6 \end{pmatrix} \begin{pmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{pmatrix} = \begin{pmatrix} 147.8 \\ 99.7 \\ 93.2 \\ 154.3 \end{pmatrix} \qquad \hat{b} = \begin{pmatrix} 9.620 \\ 0.000 \\ 18.490 \\ 20.907 \end{pmatrix}$$

$$k' = \begin{pmatrix} 1 & -1 & 0 & 0 \end{pmatrix} \qquad k'b = 9.620 \quad herd$$

$$k' = \begin{pmatrix} 0 & 0 & 1 & -1 \end{pmatrix} \qquad k'b = -2.417 \quad season$$

Equations for 2nd herd was set to zero to obtain above solutions. If 1st herd is set to zero, we get:

$$\hat{b} = \begin{pmatrix} 0.000 \\ -9.620 \\ 28.110 \\ 30.527 \end{pmatrix}$$

Note: linear contrasts (differences between levels of an effect) are the same for both solution vectors

Hypothesis testing

Is the model suitable/adequate?

Does a certain factor affect significantly a trait?

Should this factor be included in the final model of analysis?

Typical sums of squares needed:

- $SS_{total} = SST = y'Q_Ty$ with $Q_T = V^{-1}$
- $SS_{model} = SSR = y'Q_My$ with $Q_M = V^{-1}X(X'V^{-1}X)^{-1}X'V^{-1}$
- $SS_{error} = SSE = y'Q_Ey$ with $Q_E = V^{-1} - V^{-1}X(X'V^{-1}X)^{-1}X'V^{-1}$

- The ratio of two independent central chi-square variables has an F-distribution.
- Therefore the adequacy of the whole model can be tested as:
  - $F_M = SSR/ r(X) / (SSE/ (N – r(X))$
  - Where N is the number of records and r(X) is the rank of the X.

Hypothesis testing

Does season have a significant effect on milk yield?

$k'b = 30.527 - 28.110 = 2.417$

The test statistics is

- **$F = [ss / r(k')] / (SSE/(N - r(X))$**
- **Where ss = sum of squares of $k'b$ =**

$$(k'b)'(k'(X'X)^{-1}k)^{-1}k'b$$

Hypothesis testing

r(k') = 1

$$(X'X)^{-1} = \begin{pmatrix} 0.400 & 0.0 & -0.200 & -0.200 \\ 0.0 & 0.0 & 0.0 & 0.0 \\ -0.200 & 0.0 & 0.350 & 0.100 \\ -0.200 & 0.0 & 0.100 & 0.267 \end{pmatrix}$$

ss = 14.019

SSE = y'y – bX'y = 7,076.59 – 3,225.95 = 3,850.64

F = 14.019 / (3,850.64 / 7) = 0.025

- **non-significant at 5% level**

Example Data

Cow	Herd	Season	Sire	Milk Yield (kg/d)
1	1	2	1	36.2
2	1	1	2	25.8
3	1	2	3	31.5
4	1	1	1	42.0
5	1	2	2	12.3
6	2	2	3	28.5
7	2	1	1	10.6
8	2	2	2	23.4
9	2	2	3	22.4
10	2	1	1	14.8

Milk yield is a sex-controlled trait

- **phenotype is only expressed in females**
- **both females and males carry genes affecting the trait**

Fathers (sires) have multiple daughters via Artificial Insemination

Here we assess the genetic effect of the sires based on daughter phenotypes

Example Data

Cow	Herd	Season	Sire	Milk Yield (kg/d)
1	1	2	1	36.2
2	1	1	2	25.8
3	1	2	3	31.5
4	1	1	1	42.0
5	1	2	2	12.3
6	2	2	3	28.5
7	2	1	1	10.6
8	2	2	2	23.4
9	2	2	3	22.4
10	2	1	1	14.8

Sire is fitted as a random effect

In the first instance, the three sires are assumed to be unrelated to each other

$\sigma^2_e = 100$ and $\sigma^2_s = 200$

$\lambda = \sigma^2_e/\sigma^2_s = 100 / 200 = 0.50$

$$Z' = \begin{pmatrix} 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 \end{pmatrix} \quad and \quad Z'Z = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

Example MME

- The MME are

$$\begin{bmatrix} X'X & X'Z \\ Z'X & Z'Z + A^{-1}\lambda \end{bmatrix} * \begin{bmatrix} \hat{b} \\ \hat{u} \end{bmatrix} = \begin{bmatrix} X'y \\ Z'y \end{bmatrix} \qquad A = I$$

$$\begin{pmatrix} 5 & 0 & 2 & 3 & 2 & 2 & 1 \\ 0 & 5 & 2 & 3 & 2 & 1 & 2 \\ 2 & 2 & 4 & 0 & 3 & 1 & 0 \\ 3 & 3 & 0 & 6 & 1 & 2 & 3 \\ 2 & 2 & 3 & 1 & 4.5 & 0 & 0 \\ 2 & 1 & 1 & 2 & 0 & 3.5 & 0 \\ 1 & 2 & 0 & 3 & 0 & 0 & 3.5 \end{pmatrix} \begin{pmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \\ u_1 \\ u_2 \\ u_3 \end{pmatrix} = \begin{pmatrix} 147.8 \\ 99.7 \\ 93.2 \\ 154.3 \\ 103.6 \\ 61.5 \\ 82.4 \end{pmatrix}$$

Example Solutions

	σ^2_s = 200 High h^2	Fixed eff. model	σ^2_s = 5 Low h^2
Herd			
1	11.330	9.620	9.866
2	0.000	0.000	0.000
Season			
1	17.510	18.490	18.308
2	19.817	20.907	20.766
Sire			
1	1.909		0.341
2	-5.230		-0.786
3	3.320		0.445

BLUE

BLUP (PTA)

Example MME

- When random effect variance is small, the solutions for the fixed effects are more similar to those from the fixed effect model
- When random effects with large variation are present, MME present the best approach
- The estimate of λ is inferred from the data analysis with software (e.g. R, ASREML)

For the tutorial

- In a Sire Model:
  - $\lambda = (4-h^2)/h^2$
  - h^2 = var(u)/var(P) = heritability of the trait
  - var(P) = $\sigma^2 s + \sigma^2 e$
  - var(u) = $4*\sigma^2_s$
  - i.e. additive genetic variance equals four times the sire variance
    - var(u) = ¼ var(sire) + ¼ var(dam) + ½ V(Mendelian sampling)